FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY
ADVENT 2026 SEMESTER GROUP ASSIGNMENT

PROGRAM: BACHELOR OF SCIENCE IN DATA SCIENCE AND ANALYTICS

YEAR: 2

SEMESTER: 1

COURSE CODE: DSC2107

COURSE NAME: DATA MINING AND WRANGLING

ASSESSMENT TYPE: PRACTICAL GROUP ASSIGNMENT (WEB SCRAPING)

RELEASE DATE: OCTOBER 2026

DURATION: 11 DAYS (ABOUT 1.5 WEEKS)

Assignment Instructions

1. The general Uganda Christian University academic guidelines and policies apply to this assignment, including the policies on plagiarism and academic dishonesty. Violating any of them makes the group's submission void.

2. This assignment is done in our assigned groups. It is marked out of **100 marks** and contributes **30%** of the course grade.

3. Each group works **only on its assigned site** (Section 2). Swapping sites without the instructor's approval is not allowed.

4. All group work must be submitted on Moodle by one designated member of each group.

5. The **Rules of Engagement** in Section 4 are compulsory and graded. Breaking them costs marks even if the scraper works.

6. Uganda Christian University or its officials reserve the right to disqualify any submission that breaches these instructions.

1. Overview and Learning Objectives

Each group will scrape product data from one assigned real e-commerce site, store it in a database, and analyse it over a **one-week collection period**. The assignment runs for **11 days**.

By the end of the assignment, students should be able to:

(a) Inspect a live website (HTML structure and network requests) and choose the right scraping approach.

(b) Build a scraper that handles pagination, missing fields and inconsistent formatting.

(c) Collect data repeatedly on a schedule to build a time series.

(d) Clean the data and store it in a structured database or file.

(e) Analyse and visualise the data to answer real business questions.

(f) Scrape responsibly: respect `robots.txt`, limit request rates and document ethical choices.

2. Group Site Allocation

Each group is assigned one site. All 10 sites passed a readiness check on 9 October 2026: each loaded normally, `robots.txt` allowed the starting page, and product data was present. The sites are paired so groups can compare results.

Group	Site	Starting page	Data to collect
1	Jumia Kenya	`jumia.co.ke/phones-tablets` and two other categories	Name, brand, price, old price, discount %, rating, review count, category
2	Jumia Nigeria	`jumia.com.ng/phones-tablets` and the same categories as Group 1	Same fields as Group 1
3	Jumia Egypt	`jumia.com.eg/phones-tablets` and the same categories as Group 1	Same fields as Group 1
4	Kilimall Kenya	`kilimall.co.ke` category pages	Name, price, discount, rating, units sold
5	Jiji Uganda	`jiji.ug/mobile-phones` and two other categories	Title, price, condition, location, posting date (no seller names or phone numbers)
6	Jiji Kenya	`jiji.co.ke/mobile-phones` and the same categories as Group 5	Same fields as Group 5
7	Steam Top Sellers	`store.steampowered.com /search/?filter=topsellers`	Rank, title, price, original price, discount %, release date, review summary, platforms
8	itch.io Top Sellers	`itch.io/games/top-sellers`	Rank, title, developer, price, genre, platforms
9	Allbirds (Shopify)	`allbirds.com/products.json`	Product, type, tags, variants, sizes, price, availability
10	Kith (Shopify)	`kith.com/products.json`	Product, brand (vendor), type, variants, sizes, price, availability

Warm-up for all groups (Day 1): before touching their assigned site, every group scrapes all ≈ 1,000 books from `books.toscrape.com` to practise pagination and detail pages.

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3. Task Requirements

Part A: Scraping

(a) **Reconnaissance.** Inspect the site's HTML and the browser's Network tab. Decide whether to parse HTML or call the site's underlying JSON endpoints, and justify the choice.

(b) **Scraper.** Write the scraper in Python using `requests` + `BeautifulSoup`, or Playwright or Selenium where content is loaded by JavaScript. Scrapy may be used if the group prefers.

(c) **Coverage.** Collect at least **500 products per run** across at least **3 categories** (Groups 9 and 10: all products in the feed; Groups 7 and 8: the top 500 ranked items).

(d) **Robustness.** Handle pagination, missing fields, network errors (retry with back-off) and layout surprises without crashing.

(e) **Scheduling.** Run the scraper **once a day for 7 consecutive days**. Each record must carry a `scraped_at` timestamp.

Part B: Storage

(a) Save each daily run as a raw CSV or JSON file, kept unchanged as evidence.

(b) Clean the data: convert prices to numbers, strip currency symbols, parse ratings ("4.5 out of 5" $\rightarrow$ 4.5), standardise categories and remove duplicates.

(c) Load the cleaned data into a relational database (SQLite, MySQL or PostgreSQL) with at least two tables, e.g. `products` (one row per product) and `observations` (price, rank and stock per product per day).

Part C: Analysis

Every group answers the **core questions** below, then adds **one question of its own**. Analysis is done in Python (pandas, matplotlib/seaborn) in a Jupyter notebook, with at least **6 clear, labelled charts**.

(a) What does the price distribution look like in each category? Identify outliers.

(b) Is there a relationship between price and rating, or between review count and rank/popularity?

(c) How did prices, discounts and rankings change over the 7 days? Which products moved most?

(d) Which brands or sellers dominate each category?

(e) What share of products are discounted, and how large are the discounts really (old price vs new price)?

Cross-group comparisons (bonus): Groups 1, 2 and 3 compare Jumia prices across Kenya, Nigeria and Egypt in USD. Groups 1 and 4 compare Jumia and Kilimall within Kenya. Groups 5 and 6 compare second-hand phone prices in Uganda and Kenya. Groups 7 and 8 compare pricing on Steam and itch.io. Groups 9 and 10 compare pricing and stock between two fashion brands.

4. Rules of Engagement

These rules are graded. Breaking them costs marks even if the scraper works.

(a) **Check `robots.txt` first.** Record what it allows and disallows for your target pages, and stay within it.

(b) **Go slowly.** Wait at least **2–5 seconds** between requests, and cap each daily run at roughly **1,000 requests**.

(c) **Identify yourself.** Set a clear User-Agent string that names the course and a contact email, for example:
`DSC2107-Course-Project (student email)`.

(d) **No logins.** Scrape only pages visible without an account.

(e) **Public product data only.** Do not collect reviewer names, profiles, phone numbers or any other personal data. On Jiji, skip seller names and phone numbers.

(f) **No bypassing protections.** Do not solve or bypass CAPTCHAs, rotate proxies to evade blocks, or spoof identities. If you are blocked, stop, document it, and reduce your rate.

(g) **Coursework only.** Do not publish or resell the dataset.

Being blocked is not a failure. A clear write-up of what happened and how the group adjusted earns full marks for that part.

5. Deliverables

#	Deliverable	Format	Due
1	Reconnaissance note (1 page): site structure, `robots.txt` findings, chosen approach	PDF	End of Day 2
2	Working scraper + first sample run ($\geq$ 100 products)	Code repository + CSV	End of Day 3
3	Source code: scraper, cleaning and loading scripts, with a README on how to run them	GitHub repository	End of Day 10
4	Raw data (7 daily files) and the cleaned database	CSV/JSON + `.db` or SQL dump	End of Day 10
5	Analysis notebook with $\geq$ 6 charts and written interpretation	`.ipynb` + PDF export	Day 11
6	Final report (5–8 pages): method, data, findings, limitations, ethics	PDF	Day 11
7	Presentation (10 minutes + 5 minutes Q&A)	Slides	Day 11 class
8	Individual contribution statement ($\frac{1}{2}$ page per student)	PDF	Day 11

6. Timeline

The assignment runs 11 days, with daily data collection on Days 4–10. Day 1 is the day the assignment is released. Storage work overlaps with collection, so groups must split tasks.

Days	Focus	Activities	Checkpoint
1–2	Warm-up and reconnaissance	Scrape `books.toscrape.com`; inspect the assigned site; read `robots.txt`; choose tools	Reconnaissance note (Day 2)
2–3	Build the scraper	Pagination, field extraction, error handling, rate limiting; set up daily scheduling	Sample run of $\geq$ 100 products (Day 3)
4–10	Daily collection (7 runs)	One scheduled run per day; monitor failures	Instructor check of raw files on Day 7
5–8	Storage (in parallel)	Design the schema; write cleaning and loading scripts; load runs as they arrive	Database loading all runs so far
8–10	Exploration	Exploratory analysis and draft charts on partial data; finalise after the Day 10 run	Repository + database submitted (Day 10)
11	Analyse and present	Final notebook, report, presentation	All final deliverables

7. Marking Rubric [100 marks]

Group marks apply to all members; the individual contribution component can adjust a student's mark up or down.

Criterion	Marks	Excellent (90–100%)	Good (70–89%)	Satisfactory (50–69%)	Poor (<50%)
Reconnaissance and approach	10	Clear analysis of HTML and network calls; approach well justified; `robots.txt` fully documented	Sound approach with minor gaps in justification	Approach works but is poorly explained	No reconnaissance or approach unsuitable
Scraper quality	20	Handles pagination, missing fields, errors and retries; clean, modular, commented code	Works reliably with minor fragility	Works on the happy path; breaks on edge cases	Does not run or collects little data
Data collection	10	All 7 daily runs, $\geq$ 500 products each, timestamps present	6 runs or slightly below target volume	4–5 runs or large gaps	Fewer than 4 runs
Responsible scraping	10	All rules followed; rate limits and User-Agent in code; blocks documented thoughtfully	Rules followed with small lapses	Some rules ignored, e.g. no delays	Personal data collected or protections bypassed
Cleaning and storage	15	Well-designed normalised schema, data dictionary, reproducible cleaning scripts	Good schema; minor cleaning issues	Single flat table; cleaning partly manual	Data stored only as raw files
Analysis and visualisation	20	All core questions plus one original question answered with evidence; charts clear and labelled; insights go beyond description	Core questions answered well; charts mostly clear	Some questions answered superficially	Little analysis; charts missing or misleading
Report and presentation	10	Clear, well-structured; limitations and ethics discussed honestly; confident Q&A	Clear with minor gaps	Disorganised or missing sections	Report or presentation missing
Individual contribution	5	Clear, specific statement consistent with repository history	Adequate statement	Vague statement	Missing
Total	**100**				

Bonus (up to +5 marks): cross-group price comparison, an interactive dashboard, or an automated pipeline with logging and alerts.

◀ END ▶

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